

RRSL Bangalore Visit Report

Team Surge · ScaleCert · SIH26035

A test-by-test record of primary-source findings from an in-person visit to RRSL Bangalore — technician walkthrough, direct conversation with Deputy Director C. S. Sharma, and cross-checks against the OIML R 76-1:2006(E) / R 76-2:2007(E) primary text.

Location: RRSL Bangalore

Visit date: 23\september\2026

Prepared by: Team Surge

This document distinguishes, throughout, what was **directly confirmed** (by RRSL staff, the primary OIML text, or the photographed report forms) from what remains **genuinely open or unconfirmed**.

Nothing here is presented as sourced fact unless it is.

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Part A: Visit Overview

A.1 Purpose & Context

ScaleCert is Team Surge's NAWI (Non-Automatic Weighing Instrument) certification digitization tool for Smart India Hackathon 2026 (Problem Statement 26035). This visit to RRSL Bangalore was conducted to validate the real test procedures, report formats, and workflow against primary sources and direct practitioner input — rather than build further on assumptions.

A.2 Scope Correction — The Real Test Battery

Major correction, confirmed on-site: the earlier assumption that RRSL performs only four tests (Weighing, Eccentricity, Repeatability, Creep) is wrong. In reality, RRSL performs approximately **15–18 tests** on a single instrument during full type evaluation. This matches what a competing team's public materials claimed ("seventeen R 76 test forms"), now independently confirmed by direct observation, not inferred from marketing. The honest status going into this visit: 2 of ~17 real tests (Weighing, Eccentricity) implemented and validated against the primary source; Repeatability and Creep's criteria sourced and ready to implement.

A.3 Standards Obtained & Confirmed

- Physical OIML R 76-1 / R 76-2 report forms photographed directly at RRSL Bangalore.
- Full primary-text PDFs obtained: R 76-1:2006(F) and R 76-2:2007(E), extracted and cross-checked clause-by-clause throughout this walkthrough.
- **Confirmed directly by RRSL:** the lab uses the international OIML standard directly, not a separate Indian-specific schedule — resolving the longest-standing open question in the project.
- **Confirmed directly by RRSL:** the OIML R 76-2:2007 report forms photographed throughout this document are the actual, current, in-use format — not merely "the standard that theoretically applies."

A.4 Workflow & UX Findings

- **Flexible test ordering, explicitly requested by the technician:** tests are not performed in a fixed sequence in real practice (e.g. Eccentricity may be done before Creep on a given day). The system must support this, not force a linear wizard.
- **Minimal-input, auto-calculate pattern, described unprompted by Sharma** as what a good digital system should do: the technician enters only raw inputs (load, indication) and the system computes everything else (P, E, Ec, pass/fail) from R 76-1's formulas — matching ScaleCert's existing design philosophy.
- **Sharma's stated requirement:** the technician should only ever enter Indication and ΔL per row — the applied load L itself should be auto-generated by the system, not typed in manually.

Part B: Cross-Cutting Primary-Source Findings

These findings apply across multiple tests rather than to one specific form, and are grounded directly in the primary OIML R 76-1 text or RRSL's own confirmation, not inference.

B.1 The Core Change-Point Formula (Annex A.4.4.3)

$$P = I + \frac{1}{2}e - \Delta L \quad E = P - L \quad E_C = E - E_0$$

This is the base formula reused, in whole or simplified form, by nearly every test in the battery. E0 is the error calculated at or near zero, used as a zero-correction baseline — except where explicitly noted otherwise (Repeatability, Discrimination, Temperature Effect, and the other time-based tests use their own variants, detailed in Part C).

B.2 Table 3 — Classification of Instruments

R 76-1's own Table 3 (accuracy class vs. e-range vs. n-range vs. Min capacity) was obtained directly from the primary text, closing the auto-accuracy-class-determination gap in the engine. The verification scale interval rule (3.4.2: $d < e \leq 10d$, $e = 10^k$ g) was confirmed the same way.

B.3 Section 17.4 — Software-Controlled Devices

R 76-2's own Section 17.4 (clauses 5.5.1–5.5.3, G.1–G.3.7) specifies the standard's own audit-trail, checksum, and data-reconstruction requirements for software-controlled legally-relevant devices — directly paralleling ScaleCert's audit_log / hash-chain design. This is the single strongest regulatory-grounding find of the visit for that part of the system: requirements for legally-relevant software identification, checksums preventing falsified code from running, protection of stored data with checksums/parity, and an effective audit trail are all explicit in the standard itself, not just good engineering practice.

B.4 Bidirectional (↓/↑) Reading — Resolved This Session

Original text (A.4.4.1, opening sentence):

“Appliquer des charges d’essai à partir de zéro jusques et y compris Max et enlever de même les charges d’essai jusqu’à zéro.”

Translation: “Apply test loads from zero up to and including Max, and likewise remove the test loads down to zero.”

This is the base Weighing test procedure (A.4.4.1) itself — the same clause Weighing, Tare, Damp Heat, and Endurance all reference. Bidirectional testing is not an add-on seen on some forms; it is what the base procedure *is*. **Action item resolved:** the already-built Weighing engine function (CM-01) needs a signature update to support ↓/↑ readings — now a confirmed R 76-1 requirement, not a maybe. This also resolves the “10 readings” ambiguity:

10 distinct load values, each measured going up **and** coming back down (~20 total readings for the initial-intrinsic-error case), not 10 total split across directions.

B.5 Clauses 8.3.3 / 8.4 — Verification Checklist vs. Full Type Evaluation — Resolved This Session

A separate, smaller, OIML-mandated checklist exists for **Initial Verification** (clause 8.3.3), reused identically for **Subsequent Verification** (8.4.1) and **In-Service Surveillance** (8.4.2, with MPE doubled). This is the sourced origin of the already-known “in-service = 2× initial” rule.

Clause	Requirement	Refers to	Applicability note
3.5.1 / 3.5.3.3 / 3.5.3.4	Indication errors	A.4.4–A.4.6	5 loads normally sufficient (not 10); Min only if ≥ 100 mg
4.5.2 / 4.6.3	Zero-setting & tare device accuracy	A.4.2.3, A.4.6.2	—
3.6.1	Fidelity (Repeatability)	A.4.10, 3rd paragraph	—
3.6.2	Eccentricity of load	A.4.7	—
3.8	Mobility (= “Discrimination” in R76)	2A.4.8 English text)	Not applicable to digital-indication instrument
4.18	Tilting	A.5.1.3	Mobile instruments only
6.1	Sensitivity	A.4.9	Non-automatic-balance instruments only

“Other tests may be performed in special cases (very unusual construction, doubtful results, or as indicated in the corresponding OIML type-approval certificate).” — clause 8.3.3, final line.

What this means: the ~17–18 item battery this report walks through in Part C is very likely **full type evaluation**. Routine verification — probably most of RRSL's day-to-day work — requires only the 7 items above. This reframes what ScaleCert's default/most-common workflow might center on.

B.6 Load-Sequence Convention (10e / boundaries / 5 kg steps)

Confirmed by primary text (A.4.4.1): at least 10 different test loads for initial intrinsic error (5 for other weighing tests); chosen loads must include Max, Min (if $\text{Min} \geq 100$ mg), and points at or near each MPE-band transition. **The standard does NOT prescribe a starting point of “10e” nor a fixed 5 kg spacing** — that specific convention was independently confirmed by Sharma as common practice across multiple labs, developed to satisfy the standard's looser requirement, not a rule written in R 76-1 itself. Safe framing: “industry-standard lab convention for satisfying R 76-1's minimum-10-loads requirement,” not “R 76-1 requires this spacing.”

Still genuinely open: the exact intermediate load points within the first MPE band (Band 1) remain unconfirmed — an earlier placeholder (1 kg, 2 kg) was explicitly found not to match what was shown on Sharma's screen and has been discarded. The standard itself does not specify a single “correct” answer here, so this can only be resolved by direct confirmation of RRSL's own convention, not derivation.

Part C: Test-by-Test Findings (18-item battery)

Each test below follows the same structure: what the system should auto-generate, what the technician enters, what the system calculates (including the pass/fail rule), and its build status.

Test 1: Weighing Performance (A.4.4 / A.5.3.1)

System auto-generates

- Header fields from instrument registration (Application no., Type designation, e, resolution-during-test).
- Environmental conditions table (Temp / Rel.h / Time / Bar.pres at start, max, end).

Technician enters

- Observer, date.
- Zero-setting/zero-tracking device status.
- Per load: Load (L), Indication (I), Add. load (ΔL).

System calculates / pass-fail

- $E = I + \frac{1}{2}e - \Delta L - L$ $E_c = E - E_0$
- Pass/fail: $|E_c| \leq |mpe|$
- Confirmed (B.4): must be bidirectional (1/1), ≥ 10 distinct loads each direction.

BUILT — CM-01 requires bidirectional signature update (see B.4)

Test 2: Temperature Effect on No-Load Indication (A.5.3.2)

System auto-generates

- Same header as Test 1 + report-page cross-reference to the Weighing session (must be reported together per the form's footnote).

Technician enters

- Zero-setting device status.
- Per reading: Date, Time, Temp ($^{\circ}\text{C}$), Zero indication (I), Add. load (ΔL).

System calculates / pass-fail

- $P = I + \frac{1}{2}e - \Delta L$ (no $-L$ term, zero test)
- $\Delta P / \Delta \text{Temp}$ = zero-change rate between consecutive readings
- Pass: zero-change per $5^{\circ}\text{C} < e$ (Class II/III/III), per $1^{\circ}\text{C} < e$ (Class I)
- No E_c/E_0 zero-correction — a rate check, not an MPE lookup. **OUT OF SCOPE (3-day build) — needs its own evaluation function**

Test 3: Eccentricity using Weights (A.4.7 / 3.1)

System auto-generates

- Same header + environmental table as Test 1.

Technician enters

- Branching pre-question: mobile instrument? (A.4.7.5)
- Zero-setting device status.
- Per position (1=top-left, 2=top-right, 3=bottom-right, 4=bottom-left, clockwise): Load (L), Indication (I), Add. load (ΔL).

System calculates / pass-fail

- Same formula as Test 1: $E = I + \frac{1}{2}e - \Delta L - L$, $E_c = E - E_0$ • E_0 re-measured fresh before EACH position, not once globally.
- Pass/fail: $|E_c| \leq |mpe|$

BUILT (position naming corrected: 1–4 clockwise, no “center”)

System auto-generates

- Same base header.

Technician enters

- Applicability gate: “Undivided load receptor” checkbox (if checked, N/A).
- Number of sections; per section, per direction ($\rightarrow$), per location: Load, Indication, Add. load.

System calculates / pass-fail

- Same formula family as Tests 1 & 3.
- Pass/fail: $|Ec| \leq |mpe|$

OUT OF SCOPE — only applies to divided/sectioned receptors; most complex data shape seen

Test 5: Discrimination (A.4.8 (“Mobility” in R76-1 French text))

System auto-generates

- Same base header + Scale interval, d (distinct field from e).

Technician enters

- 5a Digital (A.4.8.2): Load, I1, remove load, re-add in 1/10 d steps to 1.4d, I2.
- 5b Analog (A.4.8.1): Load, I1, extra load = $|mpe|$, I2.
- 5c Non-self-indicating: Load, extra load = $0.4|mpe|$, visible-displacement Y/N.

System calculates / pass-fail

- 5a pass: $I2 - I1 \geq d$
- 5b pass: $I2 - I1 \geq 0.7 \times mpe$
- 5c pass: purely qualitative (visible displacement observed?)

Per clause 8.3.3, this test is NOT applicable to digital-indication instruments for routine verification (see B.5).

OUT OF SCOPE — three distinct procedures, none reuse existing engine logic

Test 6: Sensitivity (non-self-indicating instrument) (A.4.9)

System auto-generates

- Same base header.

Technician enters

- Load (L), extra load = $|mpe|$, measured permanent displacement (mm).

System calculates / pass-fail

- Tiered threshold by class & Max capacity:
- $\geq 1\text{mm}$ (class I/II) $\geq 2\text{mm}$ (class III/IIII, $\text{Max} \leq 30\text{kg}$) $\geq 5\text{mm}$ (class III/IIII, $\text{Max} > 30\text{kg}$)

Per clause 8.3.3, only applies to non-automatic-balance instruments.

OUT OF SCOPE — bespoke class/capacity-branched threshold, not e-multiple MPE

Test 7: Repeatability (A.4.10 / 3.6.1)

System auto-generates

- Same base header + two load values (~50% and ~100% of Max).

Technician enters

- Two independent series of 10 weighings each, at two different loads.
- Per reading: Indication (I), Add. load (ΔL).

System calculates / pass-fail

- $E = I + \frac{1}{2}e - \Delta L - L$ — NO E_c/E_0 correction (unique among tests using this formula family).
- Two criteria, checked per series: (a) $E \leq mpe$ for every individual reading; (b) $E_{max} - E_{min} \leq mpe$ across the series.

CORRECTED — *evaluateRepeatability()* needs redesign: 20 per-reading checks + 2 spread checks, not a single spread check

System auto-generates

- Same base header + timed schedule (0, 30, [35] min).

Technician enters

- P0 at 0 min (unloaded) → apply load, hold 30 min → P30 → unload → (multi-range only) wait 5 min → P35.

System calculates / pass-fail

- $P = I + \frac{1}{2}e - \Delta L$ (no $-L$ term)
- (a) $|P30 - P0| \leq 0.5e$ (b) $|P35 - P30| \leq e1$ (multi-range only)

BUILDABLE LATER — fully specified, no open sourcing questions

Test 9: Creep (A.4.11.1 / 3.9.4.1)

System auto-generates

- Same base header + fixed reading-schedule timer (0, 5, 15, 30 min, then conditionally 1h/2h/3h/4h).

Technician enters

- At each scheduled time: Indication (I), Add. load (ΔL).

System calculates / pass-fail

- $P = I + \frac{1}{2}e - \Delta L$; ΔP relative to 0-min baseline.
- (a) 30-min shortcut: $\Delta P \leq 0.5e$ at 30min AND $\Delta P(15 \rightarrow 30) \leq 0.2e$ → test ends.
- (b) 4-hour fallback (if a fails): $\Delta P \leq |mpe|$ over 4 hours.

RESOLVED & CONFIRMED — matches earlier primary-source finding exactly

Test 10: Stability of Equilibrium (A.4.12)

System auto-generates

- Same base header.

Technician enters

- 7a Printing/storage (5 trials): first printed value + 5-second min/max.
- 7b Zero-setting (5 trials): $E0 = I0 + \frac{1}{2}e - \Delta L - L0$.

- 7c Tare balancing (5 trials): same formula, tare load ~30% of Max.

System calculates / pass-fail

- 7a pass: first value within 1e of the 5-second readings.
- 7b/7c pass: $E0 \leq 0.25e$ (tightest tolerance in the battery).

OUT OF SCOPE — distinct tolerance & disturb-then-release procedure

Test 11: Tilting (A.5.1 / A.5.1.1–3)

System auto-generates

- Same base header + instrument-category selector (5 options) + limiting tilt angle.

Technician enters

- 3 load conditions (unloaded, L, Max) × 5 tilt positions (1=level reference, 2–5=tilted) — up to 15 points, each with its own zero-reference.

System calculates / pass-fail

- $E_v = I_v + \frac{1}{2}e - \Delta L_v - L$ $E_{cv} = E_v - E_{v0}$
- (a) unloaded: $|E_1 - E_{v0}|_{\max} \leq 2e$ (not valid for direct-sale/Class II)
- (b) loaded: $|E_{c1} - E_{cv}|_{\max} \leq |mpe|$ (at L and at Max)

Per clause 8.3.3, applies only to mobile instruments.

OUT OF SCOPE — most structurally complex test in the battery

System auto-generates

- Same base header.

Technician enters

- Two independent tare loads, each with a full table.
- Every column doubled into ↓ (decreasing) / ↑ (increasing) load.

System calculates / pass-fail

- Same formula as Test 1: $E = I + \frac{1}{2}e - \Delta L - L$, $E_c = E - E_0$
- Pass/fail: $|E_c| \leq |mpe|$

OUT OF SCOPE — but resolved the general ↓/↑ question (see B.4)

Test 13: Warm-Up Time (A.5.2)

System auto-generates

- Same base header + “Duration of disconnection before test” field (hours).

Technician enters

- At 0, 5, 15, 30 min after power-on: BOTH an unloaded and a loaded reading, per time point.

System calculates / pass-fail

- E_0 = error unloaded, E_L = error loaded, per time point.
- Pass/fail: $|E_L - E_0| \leq |mpe|$ at every time point.

OUT OF SCOPE — 4th distinct time-based comparison shape in the battery

Test 14: Voltage Variations (A.5.4)

System auto-generates

- Same base header + power-supply category selector (5 types).

Technician enters

- Fixed load of 10e, read at 3 voltage points: Reference, Lower limit, Upper limit.

System calculates / pass-fail

- Same formula as Test 1: $E = I + \frac{1}{2}e - \Delta L - L$, $E_c = E$
– E_0 • Pass/fail: $|E_c| \leq |mpe|$

BUILDABLE LATER — *reuses existing engine, cheap to add*

Test 15: AC Mains Voltage Dips & Short Interruptions (12.1 / Annex B.3.1)

System auto-generates

- N/A — requires physical EMC test equipment.

Technician enters

- Result per standardized disturbance condition: significant fault (Y/N), each repeated ≥ 10 times.

System calculates / pass-fail

- No calculation — purely observational Yes/No per condition. Same pattern confirmed across the whole 12.x disturbance family (12.2 electrical bursts, etc.).

HARD EXCLUSION — *software can only record results, never compute them*

System auto-generates

- Same base header — structurally a full Weighing Performance re-run as a before/after humidity baseline.

Technician enters

- Same table shape as Test 1, WITH ↓/↑ bidirectional columns present.

System calculates / pass-fail

- Same formula/pass-fail as Test 1.

OUT OF SCOPE — *but the 2nd confirmed ↓/↑ sighting (see B.4)*

Test 17: Span Stability (B.4)

System auto-generates

- N/A — a campaign-spanning meta-test, not a single session.

Technician enters

- Measurement 1 (Initial): 5 readings. Subsequent rounds tagged with cause (after temperature test / damp heat / mains disconnection / location change / other).

System calculates / pass-fail

- $E_0 = I_0 + \frac{1}{2}e - \Delta L_0 - L_0$ $E_L = I_L + \frac{1}{2}e - \Delta L - L$
- Shortcut: if $|(E_L - E_0)_{\max - \min}| \leq 0.1e$, fewer readings allowed in later rounds.
- Pass/fail: control-chart verdict — every round's average error stays inside a “maximum allowable variation” band across the whole campaign.

OUT OF SCOPE — *needs session-linking across the whole test campaign timeline*

Test 18 Endurance (A.6)

System auto-generates

- Same base header.

Technician enters

- a) Initial test (full weighing table, ↓/↑ present) → b) cycle instrument N times under a specified load → c) Final test (same table).

System calculates / pass-fail

- New column: Durability error = $|Ec_initial - Ec_final|$
- Pass/fail: durability error $\leq mpe$

OUT OF SCOPE — but the 3rd confirmed ↓/↑ sighting (see B.4)

Part D: Design Requirements for ScaleCert

D.1 Flexible Test Ordering

Confirmed directly by the technician: tests are not performed in a fixed sequence in real practice. The system must support running any applicable test in any order within a session, not a locked linear wizard.

D.2 Flexible Test Applicability (Technician-Selectable)

Not every test in the battery applies to every instrument. Some are universal (Weighing, Eccentricity-3.1, Repeatability, Creep). Others are conditional on device characteristics: Discrimination/Sensitivity differ by indication type (digital / analog / non-self-indicating); Eccentricity-3.2 requires a divided load receptor; Tilting applies only to mobile instruments; several tests are conditional on power-supply type. The full, precise rule set for every conditional test is not yet completely enumerated (see Part F).

Requirement: the system must not hard-code a fixed test battery per instrument. Instead, offer a per-session test checklist/selector, defaulting to the universal tests, with conditional tests offered but not forced — rather than trying to auto-derive applicability from device metadata the system doesn't yet have reliable rules for.

D.3 New Fields Not Yet in Schema

- Observer name, test date, resolution-during-test.
- Full environmental conditions table (start / max / end × Temp / Rel.h / Time / Bar.pres).
- Zero-setting/zero-tracking device status (per-test option sets vary).
- Free-text Remarks field, present on every form.
- Bidirectional (↓/↑) reading support in the Weighing engine (CM-01) — see B.4.

Part E: Battery Status Summary

#	Test	Clause	Status
1	Weighing Performance	A.4.4 / A.5.3.1	Correction needed
2	Temperature Effect on No-Load Indication	A.5.3.2	Out of scope
3	Eccentricity using Weights	A.4.7 / 3.1	Correction needed
4	Eccentricity using a Rolling Load	A.4.7.4 / 3.2	Out of scope
5	Discrimination	A.4.8 ("Mobility" in R76-1 French text)	Out of scope
6	Sensitivity (non-self-indicating instrument)	A.4.9	Out of scope
7	Repeatability	A.4.10 / 3.6.1	Correction needed
8	Zero Return	A.4.11.2	Buildable later
9	Creep	A.4.11.1 / 3.9.4.1	Built
10	Stability of Equilibrium	A.4.12	Out of scope
11	Tilting	A.5.1 / A.5.1.1-3	Out of scope
12	Tare (Weighing Test)	A.4.6.1	Out of scope
13	Warm-Up Time	A.5.2	Out of scope
14	Voltage Variations	A.5.4	Buildable later
15	AC Mains Voltage Dips & Short Interruptions	12.1 / Annex B.3.1	Out of scope
16	Damp Heat, Steady State	B.2 / 13a	Out of scope
17	Span Stability	B.4	Out of scope
18	Endurance	A.6	Out of scope

“Correction needed” = previously assumed complete, now confirmed to need a specific, identified change before it can be called correct.

Part F: Open Questions Still Unanswered

- The exact Band-1 intermediate load points in the auto-generated load sequence. The standard does not prescribe a specific spacing (confirmed), and RRSL's own convention for this first band specifically remains unconfirmed — only resolvable by asking Sharma/the technician directly, not derivable from the text.
- Whether the ~17–18 item battery demonstrated on-site is full type evaluation, versus RRSL's day-to-day verification work actually running the smaller, OIML-fixed 7-item checklist (clause 8.3.3). This changes what the default ScaleCert workflow should center on.
- The remaining applicability rules not covered by clause 8.3.3's 7-item list — e.g. the precise conditions for Zero Return, Warm-Up Time, Voltage Variations/Dips, Damp Heat, Span Stability, Endurance, and Stability of Equilibrium, all of which belong to full type evaluation.
- Whether the Repeatability test's two load values (~50% and ~100% of Max) follow a precise, sourced rule or are themselves a lab convention (not yet separately re-confirmed in this pass).

End of report. Source document: rrsi-visit-findings.md (Team Surge internal). All primary-source quotations are from OIML R 76-1:2006(F) and R 76-2:2007(E), obtained directly, not from any third-party or competing team's materials.